Assignment - Adding interactivity to your e-commerce project

Task description

Your task is to program a set of small and useful functions in JavaScript that are triggered by clicking on the buttons of the e-commerce project pages. For each function, you will need knowledge from the lessons of modules 2 and 3.

Purpose

The purpose of this task is to practise working with variables and basic data types such as number, string, and boolean, applying arithmetic operators and assignment operators in the context of a real e-commerce project. You also need to use comparison operators and logical operators to make simple decisions, write and call functions, add event listeners on buttons to connect functions with the user interface, and apply branch operators and loops, as well as handle exceptions, all with the goal of building functional and interactive components of the website.

Tasks

1. Define the following constants/variables:

- `PRODUCT1_NAME` (string), `PRODUCT1_PRICE` (number in RON), `PRODUCT1_QTY` (whole number ≥ 0)
- `PRODUCT2_NAME`, `PRODUCT2_PRICE`, `PRODUCT2_QTY`
- `VAT_RATE` (for example, 0.2), `CURRENCY` (string: `"USD"` or `"EUR"`), `USD_PER_EUR` (for example, 1.16)
- RAW_COUPON with the value of SAVE 10.

Show the types of these variables using typeof (at least 3 examples in the comments or by `console.log`).

2. Make a normalizeCoupon(code) function that:

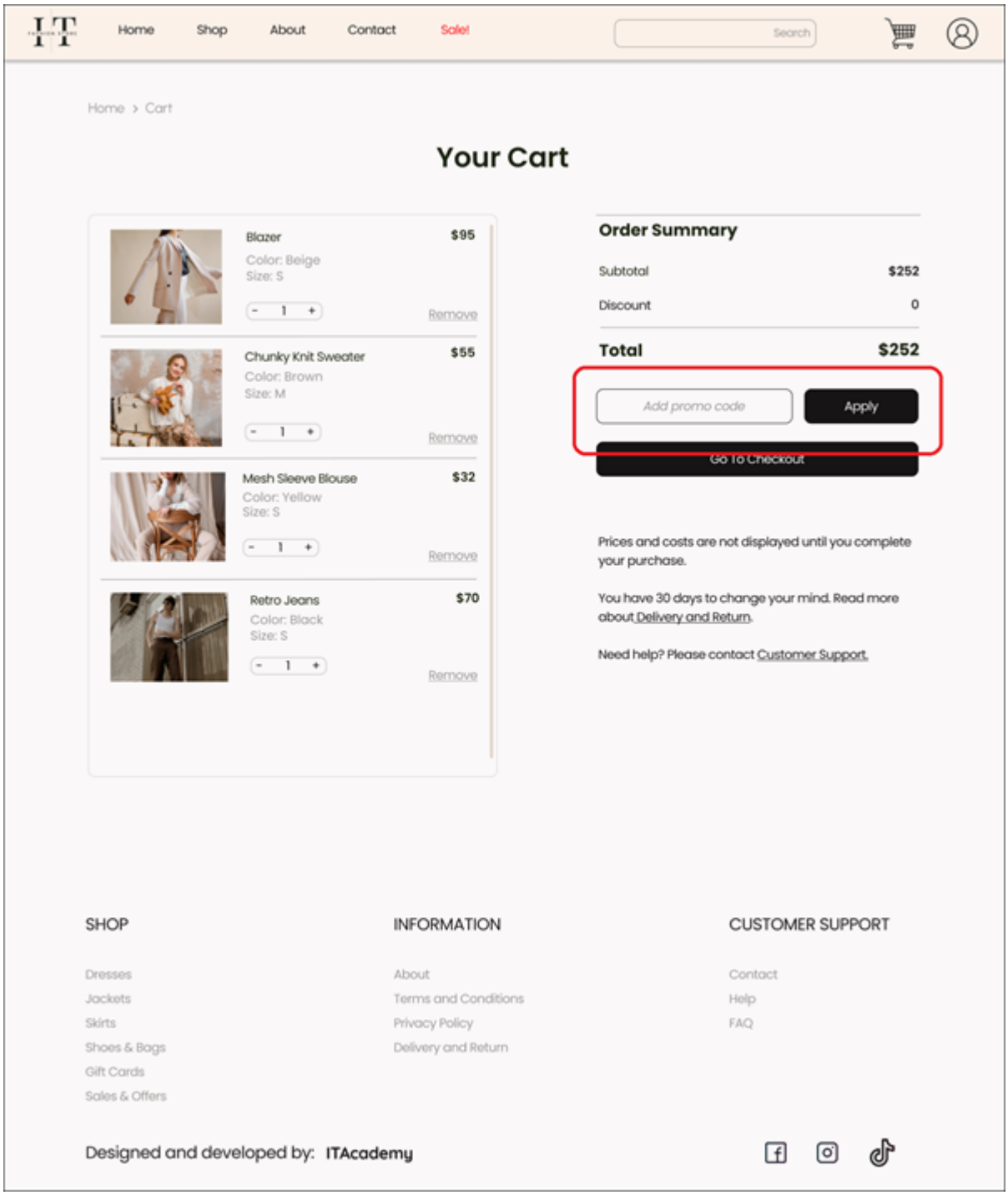
- Remove leading and trailing spaces, if any, from the value stored in the `Code` variable, using the `trim()` function.
- Converts the contents of the `code` variable to uppercase.

As a value, it returns the modified coupon.

3. In the Order Summary section of the Your Cart page, add a form with:

- an input field (placeholder: "Add promo code") and an "Apply" button.

Here's an example:



4. Coupon validation – create a new `validateAndNotify()` function that:

- Read the value in the #promo-input field.
- call normalizeCoupon(code) on this input;
- Use a condition to verify that the coupon is valid.
- Displays an alert message:
  1. If the code is valid: "The code entered is valid."
  2. if the code is not valid: "The code entered is not valid."

Note valid codes: For this exercise, consider a code to be valid if it is equal to the contents of the variable RAW_COUPON.

Do not change the order amount in this task; the purpose is only to detect and display the message to the user.

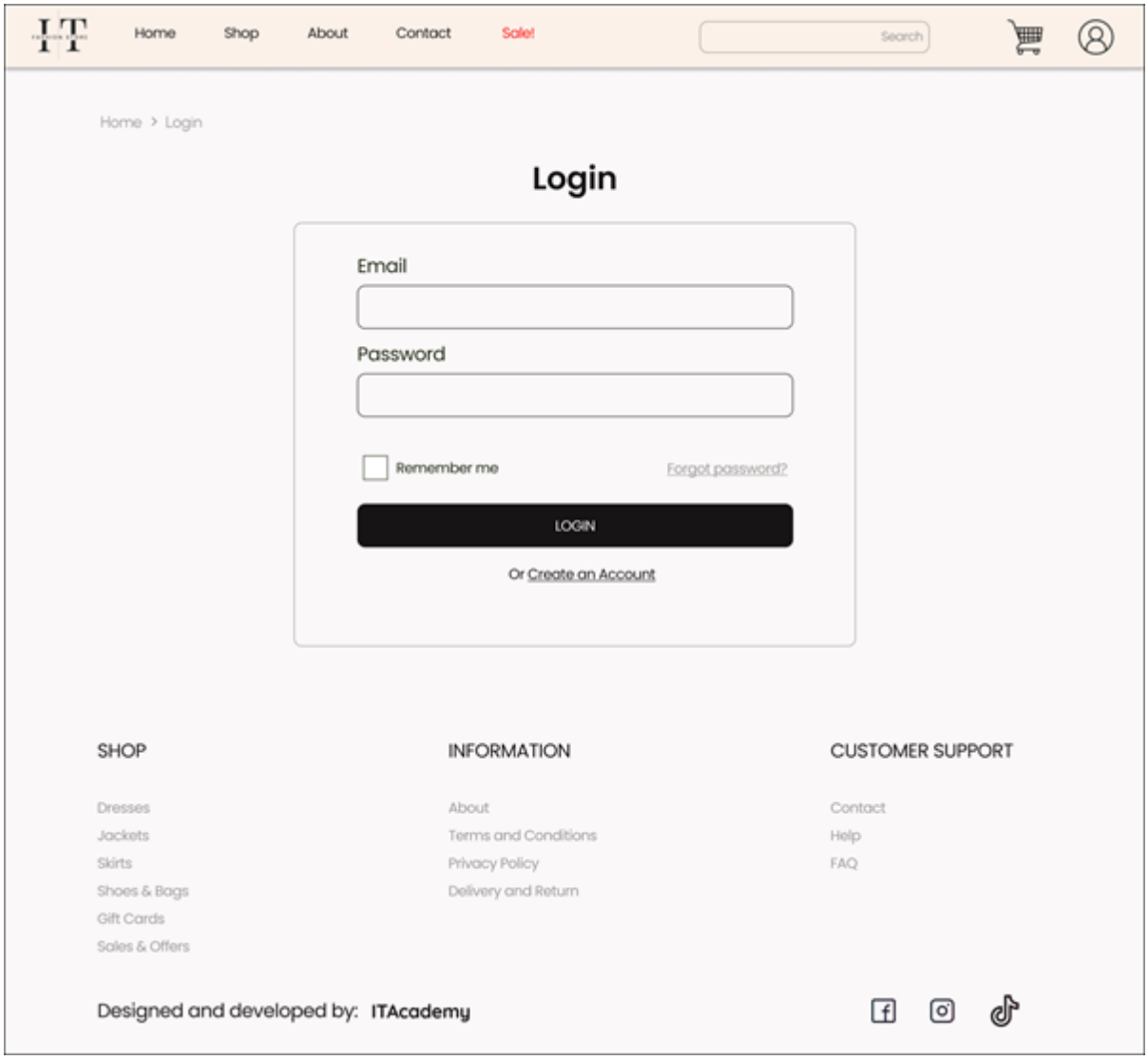
5. User Authentication – Write a `login()` function that:

- retrieves the values from the email and password fields on the HTML page;
- Remove leading and trailing spaces from both entries (using trim()).
- makes a case-sensitive comparison (it must be exactly "admin" and "admin"),
- returns true if both conditions are met, otherwise false.

Expected results (examples)

- `login("admin", "admin") → true`
- `login(" admin ", "admin") → true` (thanks to him `trim`)
- `login("Admin", "admin") → false (case-sensitive)`
- `login("admin", " ADMIN ") → false`
- `login("", "") → false`

6. Test the login feature via the HTML page.

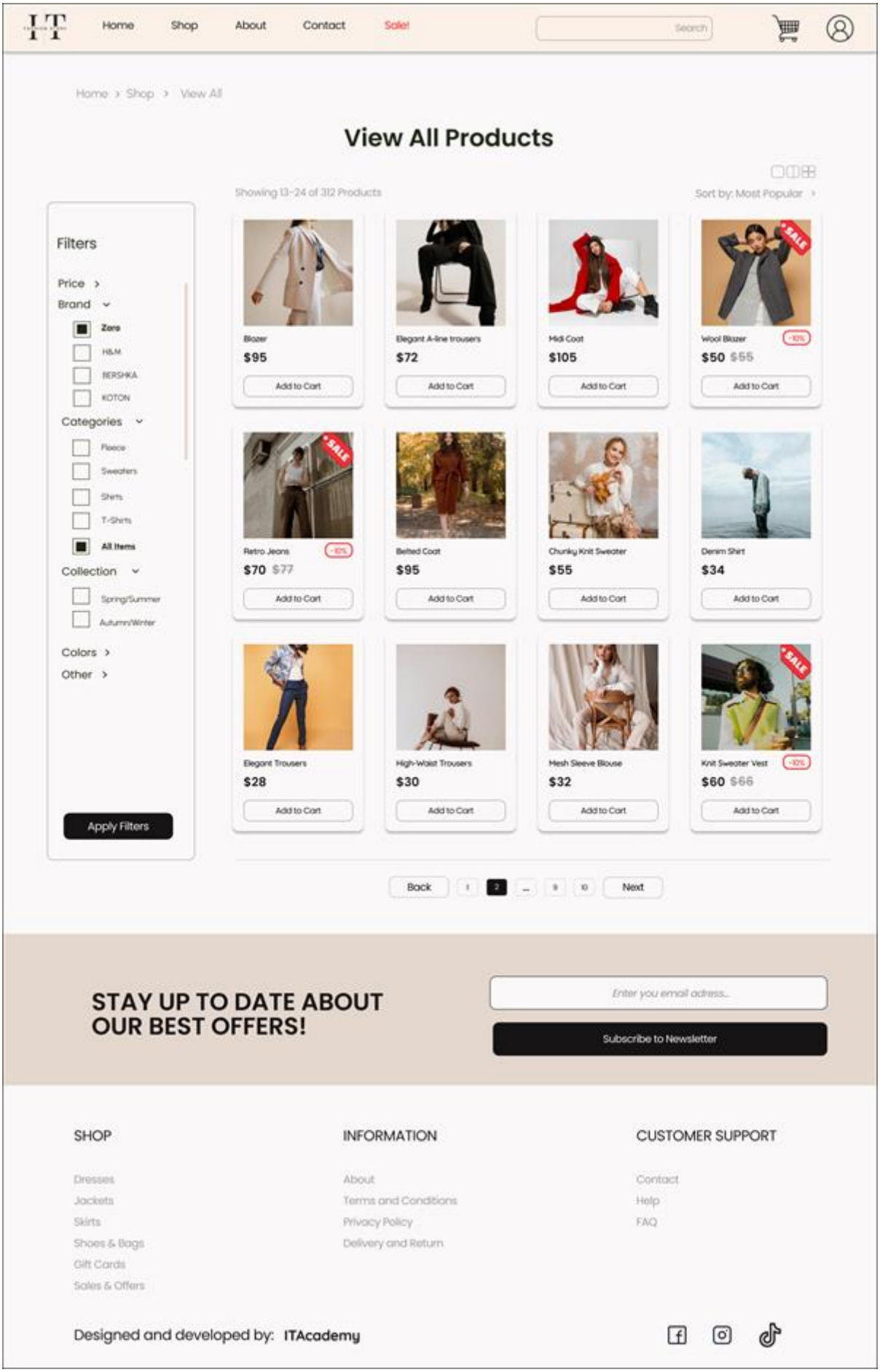


7. Track the total amount of purchases

On the View All Products page, you must maintain a global variable for the total amount of purchases in dollars. Every time the user clicks the Add to Cart button for a product, the total amount must increase with the price of that product. Your task is:

- At the top of the JavaScript file, define a global variable:
  `let suma = 0; // in $`
- Write a function that takes the price of the product as a parameter and adds it to the overall amount:
  `function adaugăLaSumă(preț) { /* ... */ }`
- Demonstrate how the feature works with at least three calls (for three different prices) and display the result after each call using `console.log`.

- Connect the function to the Add to Cart buttons in HTML using the onClick event so that with each click on the button, the function automatically increases the total amount by the corresponding product price.


Home
Shop
About
Contact
Sale!
Search
Home > Shop > View All
View All Products
Showing 13-24 of 312 Products
Sort by: Most Popular
Filters
Price
Brand
Zara
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BERSHKA
KOTON
Categories
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T-Shirts
All Items
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Spring/Summer
Autumn/Winter
Colors
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Apply Filters
Blazer
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Elegant A-line trousers
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Midi Coat
$105
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Wool Blazer
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Chunky Knit Sweater
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Denim Shirt
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Elegant Trousers
$28
Add to Cart
High-Waist Trousers
$30
Add to Cart
Mesh Sleeve Blouse
$32
Add to Cart
SALE
Knit Sweater Vest
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8. Displaying the total amount via the cart icon:

- On the site there is a cart icon in the header. Clicking on this icon should display an alert with the current total order amount (in $).

Tips for successfully performing the task

1. Structure of the function for normalizing the entered code:

```
function normalizeCoupon(code) {
   // a) elimină spațiile
   // b) transformă în majuscule
   // c) kupon returnează codul modificat
 }
```

2. If you want to get the value of an HTML element, that field must have an id. For example: `id="promo-input"`. When you want to retrieve the value of that field from JavaScript, use the command:

```
let inputCode = document. getElementById('promo-input'). value;
```

3. We add the `onClick` event as follows:

```
<button onclick="validateAndNotify()">Apply</button>
```

4. In the login function, combine the two conditions with the logical operator &.

5. To display the total amount of the order, use onClick on the cart icon that calls a function, and in the function, make an alert:

```
function openCart() {
  // alertă cu mesajul despre sumă
   }
```

Evaluation process

The course coordinator will check your paper and provide feedback within 24 hours. If you receive comments and suggestions, you'll only have one opportunity to edit and resubmit your work.

Evaluation criteria

The evaluation is based on the following essential criteria:

- Creating and Using Variables and Constants `Are` variables and constants defined and initialised correctly? Do they have significant names? Are the types of data appropriate and used correctly?
- Functions and Logic `Do` the functions (normaliseCoupon, `validateAndNotify`, login, and `addTheAmount)` work as specified? Does it return expected values and execute the expected actions?
- Validation and branching `Do` the functions correctly check the conditions (coupon validation, login)? Are branch operators and logical operators used for decision-making?
- Interactivity and user interface `Do` clicking on buttons and icons perform expected actions? Do alert messages work as described?
- Code quality and readability Is JavaScript code clean and easy to read? Are comments used to explain functions?

Sending the task

Once you have finished the assignment, upload the paper in zip format to the same page where the assignment description is located.

Save the task file as: prenume_nume_assignment02.zip

The maximum file size that can be sent in this way is 50 MB. If the file exceeds this size, use the WeTransfer website.

How to use WeTransfer

1. Open the wetransfer.com site.
2. Click on "Add your files" and select the files you want to send – you must send the file in zip format.
3. Add your email address.
4. Under Title, enter the same name as the task file.
5. In the message you must mention the name and surname and the name of the course.
6. The deadline for downloading the file is 3 days.
7. Click on the three dots and select the "Create link" option.
8. Then click on "Get a link".
9. Enter the verification code you received in the email.
10. The link obtained for download is sent to the course coordinator on the assignment submission page.